

$$* |\psi\rangle = a_1 |\psi_1\rangle + a_2 |\psi_2\rangle$$

$|a_i|^2$: Probability that a suitable measurement will find the system to be in the state $|\psi_i\rangle$

$$|\psi\rangle = \sum_n a_n |\psi_n\rangle, \quad \sum_n |a_n|^2 = 1$$

$$\langle \psi | \hat{B}^\dagger | \psi \rangle = \langle \psi | \hat{B} | \psi \rangle^*$$

$$\langle \hat{B}^\dagger \psi, \psi \rangle = \langle \psi, \hat{B} \psi \rangle = \langle \hat{B} \psi, \psi \rangle^*$$

In Dirac Notation:

$$\langle \psi | \hat{B}^\dagger | \psi \rangle^\dagger = \langle \psi | \hat{B} | \psi \rangle^*$$

Hermitian op: $\hat{A}^\dagger = \hat{A}$

$$\begin{aligned} \langle \hat{A} \rangle &\equiv \langle \hat{A} \rangle = \langle \psi | \hat{A} | \psi \rangle = \langle \psi | \hat{A} \sum_n a_n |\lambda_n\rangle \\ &= \sum_{n,m} a_m \langle \psi_m | \hat{A} a_n |\lambda_n\rangle = \sum_n \lambda_n |a_n|^2 \end{aligned}$$

$$\langle \hat{A} \rangle = \sum_n \lambda_n |a_n|^2$$

Variance:

$$\Delta A^2 \equiv \langle \psi | (\hat{A}^2 - \langle \hat{A} \rangle^2) | \psi \rangle$$

* Uncertainty $\Delta A = \sqrt{\Delta A^2}$

* Projection operator: $\hat{P} \equiv |\psi_2\rangle \langle \psi_1|$

$$\text{ex: } |\psi\rangle = a_n |\psi_n\rangle \Rightarrow \hat{P} |\psi\rangle = a_n |\psi_2\rangle \langle \psi_1 | \psi_n \rangle = a_n |\psi_2\rangle$$

• gives $|a|^2$ along $|\psi_2\rangle$.

* Any Hermitian operator can be expanded in its eigen-space Projection op's w/ coeff. eq. to their eigenvalue:

$$\hat{A} = \hat{A}^\dagger \Leftrightarrow \hat{A} = \sum_n \lambda_n \hat{P}_n$$

$$\text{where } \hat{P}_n \equiv |\lambda_n\rangle \langle \lambda_n|$$

$$\text{ex: } \hat{1} = \sum_n |\lambda_n\rangle \langle \lambda_n|$$

orth.

$$\begin{aligned}
 |\psi\rangle &= \sum_n a_n |\lambda_n\rangle \\
 \hat{A}|\psi\rangle &= \sum_m |\lambda_m\rangle \langle \lambda_m| \sum_n a_n |\lambda_n\rangle \\
 &= \sum_{m,n} a_n \langle \lambda_m | \lambda_n \rangle |\lambda_m\rangle \\
 &= \sum_n a_n |\lambda_n\rangle \\
 &= |\psi\rangle
 \end{aligned}$$

* Functions of operators: $f(\hat{A})$

$$f(\hat{A}) \equiv \sum_n f(\lambda_n) P_n$$

$$* i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$$

$$\langle A \rangle = \langle \psi(t) | \hat{A} | \psi(t) \rangle$$

$$\frac{d}{dt} \langle A \rangle = \frac{d}{dt} \langle \psi(t), A \psi(t) \rangle = \left\langle \frac{d\psi}{dt}, A \psi \right\rangle + \left\langle \psi, A \frac{d\psi}{dt} \right\rangle$$

Take S.E: $\dagger$

$$(i\hbar \frac{d}{dt} |\psi\rangle)^\dagger = (H |\psi(t)\rangle)^\dagger$$

$$-i\hbar \frac{d}{dt} \langle \psi(t) | = \langle \psi(t) | \hat{H}$$

$$\begin{aligned}
 \frac{d}{dt} \langle A \rangle &= (i\hbar \langle \psi(t) | \hat{H}) (\hat{A} |\psi\rangle) - i\hbar \langle \psi(t) | \hat{A} (\hat{H} |\psi\rangle) \\
 &= i\hbar \langle \psi(t) | [\hat{H} \hat{A} - \hat{A} \hat{H}] |\psi(t)\rangle \\
 &= i\hbar \langle \psi(t) | [\hat{H}, \hat{A}] |\psi(t)\rangle \\
 &= (-i\hbar) \langle \psi(t) | [\hat{A}, \hat{H}] |\psi(t)\rangle
 \end{aligned}$$

* $\frac{d}{dt} \langle A \rangle = 0$ implies the observable $\langle A \rangle$ is conserved.

$$|\psi\rangle = \sum_n a_n |\lambda_n\rangle \Rightarrow |\psi(t)\rangle = \sum_n a_n(t) |\lambda_n\rangle$$

$$i\hbar \sum_n \dot{a}_n |\lambda_n\rangle = \sum_n a_n \hat{H} |\lambda_n\rangle$$

$\langle \lambda_m | i\hbar \dots$

$$i\hbar \dot{a}_m = \sum_n a_n \langle \lambda_m | \hat{H} | \lambda_n \rangle$$

$$\dot{a}_m = (-i/\hbar) \sum_n a_n \langle \lambda_m | \hat{H} | \lambda_n \rangle$$

* Formal solution

$$|\dot{\psi}\rangle = \hat{H} |\psi\rangle$$

$$\begin{aligned}
 \Rightarrow \int d|\dot{\psi}\rangle / |\psi\rangle &= \int \hat{H} dt \\
 |\psi\rangle &= e^{-i/\hbar \int H dt} |\psi_0\rangle
 \end{aligned}$$

$$* |\psi(t)\rangle = U(t)|\psi(0)\rangle$$

$$\langle\psi|\psi\rangle = 1 \Leftrightarrow \langle\psi|U^\dagger U|\psi\rangle = \langle\psi|\psi\rangle$$

$$\therefore U^\dagger U = U U^\dagger = \mathbb{1}$$

Mixed States:

$$|\psi\rangle = \sum_n \sqrt{P_n} |\psi_n\rangle$$

Then the mean value of $\hat{A}$ is given by

$$\langle\psi|\hat{A}|\psi\rangle = \sum_{m,n} \sqrt{P_n} \sqrt{P_m} \langle\psi_m|\hat{A}|\psi_n\rangle$$

* Introduce the Density operator / Matrix: $\hat{\rho}$

$$\hat{\rho} = \sum_n P_n |\psi_n\rangle \langle\psi_n|$$

Weighted sum of projection operators:

$$\hat{\rho} = \sum_n P_n \hat{P}_n$$

* The Density operator for which

$$\hat{\rho} = |\psi_m\rangle \langle\psi_m|$$

is called a Pure State density operator

Mixed State $\hat{\rho} : \sum_n P_n |\psi_n\rangle \langle\psi_n|$
 "statistical mixture"

$$\hat{\rho}^\dagger = \hat{\rho}$$

Note: $\langle\phi|\hat{\rho}|\phi\rangle = \langle\phi|(\sum_n P_n |\psi_n\rangle \langle\psi_n|)|\phi\rangle$
 $= \sum_n P_n \langle\phi|\psi_n\rangle \langle\psi_n|\phi\rangle$
 $= \sum_n P_n \langle\phi|\psi_n\rangle \langle\phi|\psi_n\rangle^*$
 $= \sum_n P_n |\langle\phi|\psi_n\rangle|^2 \geq 0$

* Also, $\hat{\rho}$ can be diagonalized:

$$\hat{\rho} = \sum_m p_m |p_m\rangle \langle p_m|$$

p_m : eigenvalues

Recall $\text{Tr}(A) \equiv \sum_n \langle\lambda_n|A|\lambda_n\rangle$

and so

$$\text{Tr}(\hat{\rho}\hat{A}) = \sum_m \langle\lambda_m|(\sum_n P_n |\psi_n\rangle \langle\psi_n|)\hat{A}|\lambda_m\rangle$$

$$= \sum_n P_n \langle \psi_n | \hat{A} | \psi_n \rangle$$

$$\boxed{\text{Tr}(\hat{\rho} \hat{A}) = \langle \hat{A} \rangle}$$

$$* \text{Tr}(\hat{\rho}) = \sum_n P_n = 1$$

$$\begin{aligned} \text{Tr}(\hat{\rho}) &= \sum_n \langle \psi_n | \left(\sum_m P_m | \psi_m \rangle \langle \psi_m | \right) | \psi_n \rangle \\ &= \sum_n P_n = 1 \end{aligned}$$

likewise

$$\text{Tr}(\hat{\rho}) = \sum_n p_n = 1$$

$$\begin{aligned} * \text{Tr}(\hat{\rho}^2) &= \sum_n \langle p_n | \left(\sum_m p_m | p_m \rangle \langle p_m | \right) \left(\sum_l p_l | p_l \rangle \langle p_l | \right) | p_n \rangle \\ &= \sum_{n,l,m} \delta_{nm} p_m \delta_{ml} p_l \delta_{ln} \end{aligned}$$

$$= \sum_n p_n^2$$

$$\text{Note: } \sum_n p_n^2 \leq \sum_n \sum_m p_n p_m$$

Proof:

$$\sum_l p_l = 1 \quad \text{thus}$$

$$p_l \leq 1 \quad \forall l$$

$$\therefore \text{If } p_l \neq 1 \quad \forall l$$

$$p_l < 1$$

$$p_l^2 < 1$$

$$0 < p_l < 1$$

$$\Rightarrow 0 < p_l^2 < p_l < 1$$

$$\therefore \sum_l p_l^2 \leq \left(\sum_l p_l \right) \cdot 1$$

$$\leq \left(\sum_{l,m} p_l p_m \right)$$

$$\sum_l p_l^2 \leq 1$$

Thus $\text{Tr}(\hat{\rho}^2) \leq 1$

Claim: $\text{Tr}(\hat{\rho}^2) = 1$ Pure state
 $\text{Tr}(\hat{\rho}^2) < 1$ Mixed state

proof: If $\hat{\rho}$ is pure, then

$$\hat{\rho} = |\psi_m\rangle\langle\psi_m|$$

$$\begin{aligned}\text{Tr}(\hat{\rho}^2) &= \sum_e \langle\psi_e| (|\psi_m\rangle\langle\psi_m| |\psi_m\rangle\langle\psi_m|) |\psi_e\rangle \\ &= \sum_e |\langle\psi_e|\psi_m\rangle|^2 \\ &= 1\end{aligned}$$

(ii) If $\hat{\rho} = \sum P_n P_n$

$$\begin{aligned}\text{Tr}(\hat{\rho}^2) &= \sum_e \langle\psi_e| \left(\sum_n P_n P_n \right) \left(\sum_m P_m P_m \right) |\psi_e\rangle \\ &= \sum_e \langle\psi_e| \sum_n P_n P_m P_n |\psi_m\rangle \\ &= \sum_{n,m} P_n P_m \delta_{me} \delta_{en} \langle\psi_n|\psi_m\rangle \\ &= \sum_{n,m} P_n P_m \delta_{nm} \langle\psi_n|\psi_n\rangle \\ &= \sum_n P_n^2 \leq 1\end{aligned}$$

$\text{Tr}(\hat{\rho}^2)$: Purity of the state.
 Define the matrix elements:

$$\rho_{nm} \equiv \langle\lambda_n|\hat{\rho}|\lambda_m\rangle = \rho_{mn}^*$$

* Evolution of the Density operator:

$$\hat{\rho}(0) = \sum_n P_n |\psi_n(0)\rangle\langle\psi_n(0)|$$

$$i\hbar \frac{d}{dt} |\psi_0\rangle = \hat{H} |\psi_0\rangle$$

$$i\hbar \frac{d}{dt} |\psi_0\rangle = U^\dagger(t) \hat{H} U(t) |\psi_0\rangle$$

$$= \frac{-i}{\hbar} \hat{H}$$

$$\langle \psi(t) | U = \langle \psi(0) |$$

$$\therefore \rho(0) = \sum_n P_n U^\dagger |\psi_n\rangle \langle \psi_n| U$$

$$\therefore U \rho(0) U^\dagger = \rho(t)$$

$$\rho(t) = U(t) \rho(0) U^\dagger(t)$$

Time evolution of the density op:

$$\frac{d}{dt} \hat{\rho}(t) = -\frac{i}{\hbar} H \hat{U}(t) \rho(0) U^\dagger + \frac{i}{\hbar} U(t) \rho(0) H U^\dagger$$

$$= -\frac{i}{\hbar} \hat{U}(t) \{ \hat{H}, \rho(0) \} \hat{U}^\dagger(t)$$

Note $[U, H] = 0$ and so

$$= -\frac{i}{\hbar} [\hat{H}, \rho(t)]$$

$$\therefore \frac{d}{dt} \hat{\rho}(t) = -\frac{i}{\hbar} [\hat{H}, \rho(t)]$$

* Density op evolves in Sch. picture since $\hat{\rho}$ is expanded in $|\psi_t\rangle$.

- All states $|\psi_n\rangle$ are prepared w/ probability P_n
 $\rightarrow$ A single density operator may represent many distinct ensembles of prepared states and probabilities

ex: $\langle 0|1\rangle = 0 \Rightarrow \langle 0|0\rangle = \langle 1|1\rangle = 1$

$$\hat{\rho}_1 = \frac{1}{2} |0\rangle \langle 0| + \frac{1}{2} |1\rangle \langle 1|$$

v.s.

$$\rho_2 = \frac{1}{3} |0\rangle \langle 0| + \frac{1}{3} \frac{(|0\rangle + \sqrt{3} |1\rangle) (\langle 0| + \sqrt{3} \langle 1|)}{2}$$

$$+ \frac{1}{3} \frac{1}{2} \frac{1}{2} (|0\rangle - \sqrt{3} |1\rangle) (\langle 0| - \sqrt{3} \langle 1|)$$

- A density operator

$$\hat{\rho} = \sum_e Q_e |\psi_e\rangle \langle \psi_e|$$

$|\psi_e\rangle$ are prepared w/ prob. Q_e . This is equal to

$$\hat{\rho} = \sum_m P_m |\psi_m\rangle \langle \psi_m|$$

$$\sum_e Q_e |\phi_e\rangle \langle \phi_e| = \sum_m P_m |\psi_m\rangle \langle \psi_m|$$

$$* \sum_e \langle \psi_n | \phi_e \rangle Q_e |\phi_e\rangle = \sum_m P_m \langle \psi_n | \psi_m \rangle |\psi_m\rangle$$

$$\sum_e u_{ne} Q_e |\phi_e\rangle = P_n |\psi_n\rangle$$

* For $\{|\lambda_n\rangle\}$, $\{|\phi_m\rangle\}$ and $\hat{A}$ acting on the first Hilbert space $\mathcal{H}_1 \otimes \mathcal{H}_2 \ni \psi$

$$|\psi\rangle = \sum_n a_n |\lambda_n\rangle \otimes |\phi_n\rangle$$

$\uparrow \quad \quad \uparrow$
 $\mathcal{H}_1 \quad \mathcal{H}_2$

$$\hat{A}_1 \equiv \hat{A} \otimes \mathbb{I}_2$$

$$\langle \psi | \hat{A}_1 | \psi \rangle = \sum_m a_m^* \langle \lambda_m | \otimes \langle \phi_m | (\hat{A}_1 \otimes \mathbb{I}_2) \sum_n a_n |\lambda_n\rangle \otimes |\phi_n\rangle$$

$$= \sum_{m,n} a_n^* a_m \langle \phi_n | \mathbb{I}_2 | \phi_m \rangle \langle \lambda_n | \hat{A}_1 | \lambda_m \rangle$$

$$= \sum_{m,n} a_n^* a_m \delta_{nm} \langle \lambda_n | \hat{A}_1 | \lambda_m \rangle$$

$$= \sum_m |a_m|^2 \langle \lambda_m | \hat{A}_1 | \lambda_m \rangle$$

$$\langle \hat{A} \rangle = \text{Tr}(\hat{A} \otimes \mathbb{I}_2 \hat{\rho})$$

Here it appears, we are replacing $\hat{\rho}$ by $|\psi\rangle \langle \psi| \dots$ not sure why.

Qubits: a quantum system having two distinct (i.e. orthogonal) states. $\{|0\rangle, |1\rangle\}$
 $\langle 0 | 1 \rangle = 0$

* General Qubit: $|\psi\rangle = a_0 |0\rangle + a_1 |1\rangle$

ex: (i) Two orthogonal polarization states of a photon $\hat{A}_\mu$

(ii) Orientation of a spin-half particle

(iii) Pair of electronic energy levels of an atom, ion, or quantum dot

(iv) Paths in an interferometer.

→ Take (ii) Approach of a spin $1/2$ particle

$$|0\rangle \langle 0| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$|1\rangle\langle 1| = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes (0,1) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\hat{1} = |0\rangle\langle 0| + |1\rangle\langle 1|$$

$$* |0\rangle\langle 1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes (0,1) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$|1\rangle\langle 0| = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes (1,0) = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$\hat{\sigma}_x = |0\rangle\langle 1| + |1\rangle\langle 0| = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\hat{\sigma}_y = i \left(\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right) = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\sigma_x^2 =$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \hat{1}$$

$$\sigma_x^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Thus

$$\hat{1} = |0\rangle\langle 0| + |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\hat{\sigma}_x = |0\rangle\langle 1| + |1\rangle\langle 0| = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\hat{\sigma}_y = i \left(|1\rangle\langle 0| - |0\rangle\langle 1| \right) = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\hat{\sigma}_z = |0\rangle\langle 0| - |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

→ $\hat{\sigma}_x$: x-component of the Angular Momentum

$\hat{\sigma}_y$: y-...

$\hat{\sigma}_z$: z-...

$$\text{ex: } \hat{\sigma}_x \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} a_1 \\ a_0 \end{pmatrix}$$

Q: how would this give the Angular no.

$$S_i \equiv \frac{1}{2} \epsilon_{ijk} \sigma_j \sigma_k \leftarrow (\text{I think})$$

$\text{SU}(2)$: Algebra

$$[\hat{\sigma}_x, \hat{\sigma}_y] = 2i\hat{\sigma}_z \quad \left| \quad \{\hat{\sigma}_i, \hat{\sigma}_i\} = 0 \right.$$

$$[\hat{\sigma}_y, \hat{\sigma}_z] = 2i\hat{\sigma}_x$$

$$[\hat{\sigma}_z, \hat{\sigma}_x] = 2i\hat{\sigma}_y$$

* $[,] \neq 0$ implies that these operators have different eigenvectors.

eigenvalues: $\pm 1 \quad \forall$

eigenvectors:

Eigenvectors:

$$\sigma_x: (|0\rangle \pm |1\rangle)/\sqrt{2}$$

$$\begin{aligned} \text{Proof: } (|0\rangle\langle 1| + |1\rangle\langle 0|)(|0\rangle \pm |1\rangle)/\sqrt{2} \\ = \frac{1}{\sqrt{2}}|1\rangle \pm \frac{1}{\sqrt{2}}|0\rangle = \frac{1}{\sqrt{2}}(|1\rangle \pm |0\rangle) \\ = \pm () \quad \checkmark \end{aligned}$$

$$\sigma_y: (|0\rangle \pm i|1\rangle)/\sqrt{2}$$

$$\sigma_z: |0\rangle \text{ and } |1\rangle$$

* Note, Since $|\psi\rangle = a_0|0\rangle + a_1|1\rangle$ is
 subject to $\langle\psi|\psi\rangle = \|\psi\|^2 = 1$
 $\Rightarrow a_0^2 + a_1^2 = 1$ Then the qubit
 is subject to lying on S^2

Bloch Sphere:

• Every point on the Bloch sphere corresponds to a
 Different pure state.

→ Note on S^2 : Since $|\psi\rangle$ defines the
 state of a spin $1/2$ particle w/
 $|0\rangle$ is $+1/2$
 $|1\rangle$ is $-1/2$

along the z-axis

→ Further more:

To find the appropriate embedding we
 should look at eigenstates of σ_i
 since $|0\rangle$ and $|1\rangle$ are eigenstates
 of $\hat{\sigma}_z$:

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Likewise:

To find the " $\hat{x}$ " vector we look
 at the E. States of σ_x :

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Rightarrow (|0\rangle \pm |1\rangle)/\sqrt{2}$$

$$\text{Note: } \sigma_x (|0\rangle \pm |1\rangle)/\sqrt{2}$$

$$|\hat{x}\rangle \equiv \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = |\hat{x}\rangle$$

• Now we have

$$\sigma_y = i(|1\rangle\langle 0| - |0\rangle\langle 1|) = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

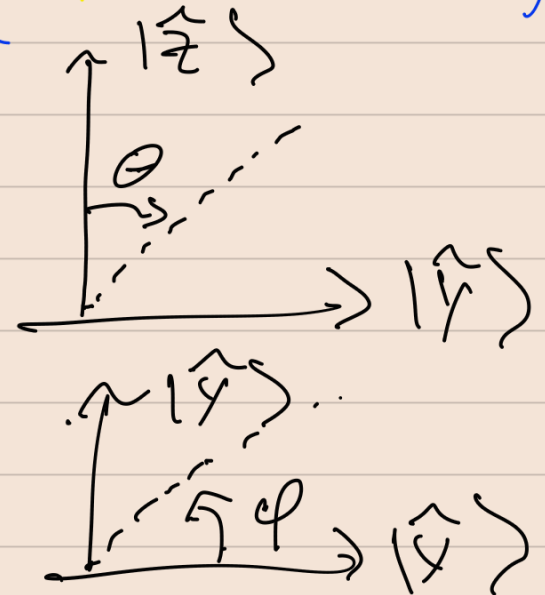
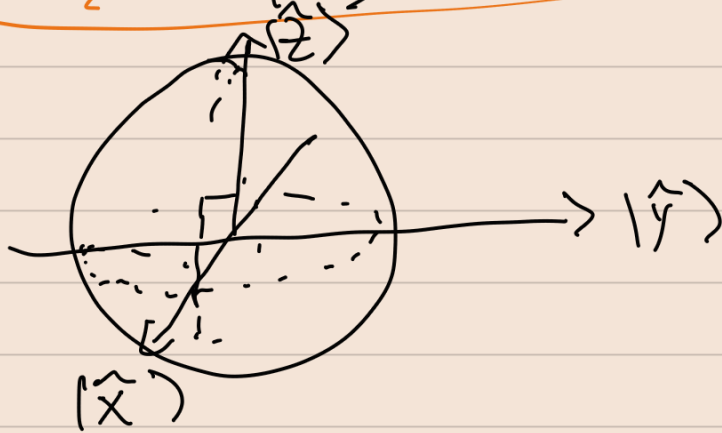
$$\sigma_y (|0\rangle + i|1\rangle) / \sqrt{2} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ i \end{pmatrix} = \begin{pmatrix} 1 \\ i \end{pmatrix} \quad \text{if } +$$

$$= \begin{pmatrix} 1 \\ i \end{pmatrix} \quad \checkmark$$

$$\therefore |\hat{y}\rangle \equiv \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle)$$

$$\begin{aligned} |\hat{x}\rangle &\equiv \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \\ |\hat{y}\rangle &\equiv \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle) \\ |\hat{z}\rangle &\equiv |0\rangle \end{aligned}$$

(Basis for Bloch Sphere)



* A general state will be given by

$$\begin{aligned} |4\rangle &= a_x |\hat{x}\rangle + a_y |\hat{y}\rangle + a_z |\hat{z}\rangle \\ &= a_x \left(\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right) + a_y \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle) + a_z |0\rangle \\ &= \left[a_z + \frac{1}{\sqrt{2}} (a_y + a_x) \right] |0\rangle + \frac{1}{\sqrt{2}} (a_x + i a_y) |1\rangle \end{aligned}$$

$$a_x^2 + a_y^2 + a_z^2 = 1$$

* Spherical Coordinates:

$$\begin{aligned} x &= \sin(\theta) \cos(\phi) \\ y &= \sin(\theta) \sin(\phi) \\ z &= \cos(\theta) \\ a_x &= \sin(\theta) \cos(\phi) \\ a_y &= \cos(\theta) \sin(\phi) \\ a_z &= \cos(\theta) \end{aligned}$$

$$\frac{1}{\sqrt{2}} (a_x + i a_y) = \frac{1}{\sqrt{2}} (\sin(\theta) \cos(\phi) + i \cos(\theta) \sin(\phi))$$

$$\frac{1}{\sqrt{2}} (\sin(\theta) \cos(\phi) + i \sin(\theta) \sin(\phi))$$

$$= \frac{1}{\sqrt{2}} \sin(\theta) e^{i\phi}$$

not Right for $|2\rangle$ but not quite for $|0\rangle$

$\therefore$ General state on the Bloch sphere is given by

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right)|1\rangle$$

* Any Unitary operator can be written as

$$\hat{U} = \exp(i\alpha \hat{I} + i\beta \vec{a} \cdot \vec{\sigma})$$

i.e. Since $\hat{U} \in SO(4,3)$ group it is expanded in terms of its generators $\{\hat{I}, \sigma_i\}$ of $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$

where $\alpha, \beta \in \mathbb{R} \mid |\vec{a}|^2 = 1 = a_x^2 + a_y^2 + a_z^2$

Claim: We can expand $\hat{U}$ as follows

$$\hat{U} = e^{i\alpha} (\cos(\beta) \hat{I} + i \sin(\beta) \vec{a} \cdot \vec{\sigma})$$

Proof: Recall

$$f(\hat{A}) = \sum_n f(|\lambda_n|) |\lambda_n\rangle \langle \lambda_n|$$

or Taylor expand

$$f(\hat{A}) = \mathbb{I} f(0) + \hat{A} f'(0) + \frac{\hat{A}^2}{2} f''(0) + \dots$$

$$= e^{i\alpha} \mathbb{I} e^{i\beta \vec{a} \cdot \vec{\sigma}}$$

$$\text{expand } e^{i\beta \vec{a} \cdot \vec{\sigma}}$$

$$\mathbb{I} + i\beta \vec{a} \cdot \vec{\sigma} + \frac{1}{2} \beta^2 (\vec{a} \cdot \vec{\sigma})^2 + i^3 \beta^3 (\vec{a} \cdot \vec{\sigma})^3 + \dots$$

$$\Rightarrow a_i \sigma_i a_j \sigma_j = a_i a_j \frac{1}{2} \{\sigma_i, \sigma_j\} = \delta_{ij} a_i a_j$$

$$= \mathbb{I} + i\beta \vec{a} \cdot \vec{\sigma} + \frac{1}{2} \beta^2 \delta_{ij} a_i a_j +$$

$$\text{But } (\vec{a} \cdot \vec{\sigma})^2 = \mathbb{I} \quad \forall \vec{a}$$

$$\mathbb{I} + i\beta \vec{a} \cdot \vec{\sigma} - \frac{1}{2} \beta^2 \mathbb{I} - \frac{i}{3} \beta^3 (\vec{a} \cdot \vec{\sigma}) + \frac{1}{4} \beta^4 \mathbb{I} + \dots$$

$$= e^{i\alpha} (\cos(\beta) \mathbb{I} + i \sin(\beta) \vec{a} \cdot \vec{\sigma})$$

• α acts to change the phase of the state vector
Recall, the generators span a basis of $\mathfrak{g}$
and so any element of $SU(2) \otimes \overline{SU(2)}$

$$\rho = \frac{1}{2} (1 + u_1 \hat{\sigma}_x + u_2 \hat{\sigma}_y + u_3 \hat{\sigma}_z)$$

$\Rightarrow \text{tr}(\rho) = 1$ $u_i \in \mathbb{R}$